MATH 101

Chapter 6 Homework

Exponential and logarithmic functions

Sections 6.1-6.6 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Give exact answers. Use a basic four-function calculator only if needed; show algebraic reasoning.

Exponential functions and inverse language

1. Evaluate f(-1), f(0), and f(2). State the growth factor and percent increase for each unit increase in x.

$f(x) = 3 \cdot 2^x$

2. On the same axes sketch f and g using x = -1, 0, 1, 2. Label intercepts and identify which function grows faster.

$f(x) = 2 \cdot 2^x, \qquad g(x) = 2 \cdot 3^x$

3. Sketch the function. Give its domain, range, horizontal asymptote, and end behavior.

$h(x) = 3\left(\frac{1}{2}\right)^x$

4. Rewrite each statement in the other form (logarithmic or exponential).

a) $2^5 = 32$ b) $\log_3\left(\frac{1}{9}\right) = -2$

5. Evaluate exactly.

$\log_2 8, \quad \log_3\left(\frac{1}{27}\right), \quad \log_5 1, \quad \ln(e^4)$

Logarithmic graphs and properties

6. Give the domain, range, vertical asymptote, x-intercept, and end behavior. Sketch using two convenient points.

$f(x) = \log_2(x - 3) + 1$

7. Use logarithmic properties to evaluate exactly. Here log means base 10.

a) $\log 4 + \log 25$ b) $\log 45 - \log 0.45$

8. Expand as a sum or difference of logarithms. Assume x and y are positive.

$\ln\left(\frac{x^2\sqrt{y}}{3}\right)$

9. Condense to one logarithm, and state the original domain.

$2\log_3 x - \log_3(x - 1)$

10. Use change of base to evaluate exactly.

$$\frac{\log 16}{\log 4}$$

Equations and domain checks

11. Solve by rewriting with a common base.

$$8^{x-1} = 4^{x+1}$$

12. Solve exactly using logarithms; no decimal approximation is required.

$$5^{x} = 7$$

13. State domain restrictions and solve. Check the candidate.

$$\log_2(x-1) + \log_2(x+1) = 3$$

14. State domain restrictions and solve.

$$\log_3(x+4) - \log_3 x = 1$$

15. A student claims ln(a + b) = ln(a) + ln(b) for positive a and b. Use a = b = 1 to disprove the claim. State the correct product property.